
Design and Implementation of a Face Recognition System for 10 Individuals

Abstract

Face recognition has become an important area of research in computer vision and artificial intelligence due to its wide applications in security, authentication, and human-computer interaction. This project presents the design and implementation of a face recognition system capable of identifying 10 individuals with high accuracy. The system integrates face detection, preprocessing, feature extraction, and classification stages. Principal Component Analysis (PCA) was employed for dimensionality reduction, while classification was performed using K-Nearest Neighbors (KNN) and Support Vector Machine (SVM). Experimental evaluation demonstrated promising accuracy across test datasets. The results highlight the feasibility of deploying the proposed system in controlled environments and

provide insights for future improvements.

Keywords— Face Recognition, PCA, KNN, SVM, Machine Learning, Computer Vision.

I. Introduction

Face recognition is a biometric technology that has gained increasing importance due to its non-intrusive and natural way of identifying individuals. Unlike fingerprint or iris-based systems, face recognition does not require active participation from users, making it suitable for surveillance and real-time applications [1].

The objective of this project is to implement a face recognition system for a dataset containing images of 10 individuals. The system aims to detect, process, and classify faces using machine learning methods.

The contributions of this work include:

- Designing a preprocessing pipeline to prepare facial images for feature extraction.
- Implementing PCA for dimensionality reduction.
- Comparing KNN and SVM classifiers in terms of recognition performance.

The rest of this paper is organized as follows: Section II discusses related

work. Section III presents the methodology, including dataset preparation, preprocessing, and algorithms. Section IV describes the experimental setup and results. Section V concludes the paper and suggests future work.

II. Related Work

Face recognition has been extensively studied over the last three decades. Early methods focused on geometric feature-based approaches [2], while more recent works emphasize machine learning and deep learning techniques [3]. PCA, also known as Eigenfaces, was one of the pioneering methods for face recognition [4]. Although modern approaches rely on deep neural networks [5], PCA combined with traditional classifiers such as KNN and SVM remains relevant for small-scale systems and academic projects due to their simplicity and interpretability.

This project builds upon these classical methods to demonstrate their applicability in recognizing a limited set of individuals.

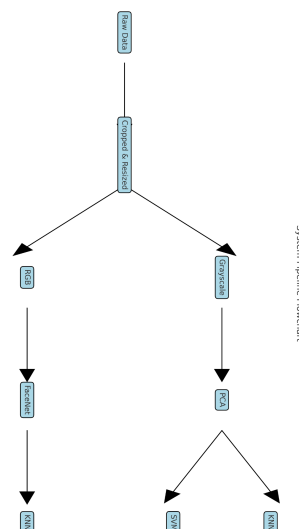
III. Methodology

A. Dataset

The dataset used in this project was created by the authors, as no well-defined and suitable dataset was readily available for the specific scope of recognizing 10 individuals. To build the dataset, photographs of 10 participants were captured under varying conditions, including different poses, facial expressions, and lighting environments. This ensured diversity within the dataset and helped improve the robustness of the recognition system.

Each image was then preprocessed by resizing it to 64×64 pixels and converting it to grayscale for consistency and reduced computational complexity. The final dataset thus provided a controlled yet diverse set of facial images, enabling effective training and evaluation of the proposed recognition system.

B. Preprocessing



Preprocessing plays a crucial role in ensuring that the input data is clean, consistent, and suitable for feature extraction. In this project, several preprocessing techniques were applied to improve the quality of the dataset:

- **Face Detection and Alignment:**

Using Multi-task Cascaded Convolutional Networks (MTCNN), facial regions were detected, aligned, and cropped. To ensure consistency, an additional margin of approximately 25% around each face was included. This step helped normalize variations in pose and position, thereby making the dataset more well-defined for recognition tasks.

- **Normalization:** Pixel values were normalized to enhance model training stability and ensure uniform scaling across all samples.

- **Resizing & Color**

Transformation: Images were resized to 64×64 pixels. Two different preprocessing pipelines were followed depending on the model:

1. **Grayscale Conversion +**

PCA: Images were converted to grayscale, reducing the data to a single channel. These were then transformed using Principal Component Analysis (PCA) for dimensionality reduction.

2. **RGB Embeddings +**

FaceNet: In an alternative approach, the images were kept in RGB format and embedded using FaceNet to extract high-dimensional feature vectors. These embeddings were later used to evaluate similarity through distance metrics and to compare recognition performance.

This multi-approach preprocessing allowed the system to evaluate both classical methods (PCA + KNN/SVM) and deep-learning-inspired embeddings (FaceNet), providing a comprehensive understanding of the trade-offs in performance.

C. Models Used for Training

In this project, two classical machine learning models — **K-Nearest Neighbors (KNN)** and **Support Vector Machine (SVM)** — were selected for training and evaluation. Both algorithms have distinct characteristics that make them suitable for small-to-medium sized datasets, such as the one constructed in this study.

1. K-Nearest Neighbors (KNN):

KNN is a simple, non-parametric classification algorithm that assigns a label to a new input sample based on the majority class of its nearest neighbors in the feature space. The distance metric, typically Euclidean distance, determines the closeness of samples.

○ Reason for Selection:

KNN was chosen because of its straightforward implementation and interpretability. It does not assume any underlying data distribution, making it flexible for facial datasets where patterns are not strictly linear. Since the dataset contains only 10 individuals, KNN is

computationally efficient and suitable for baseline comparisons.

○ Effect on Dataset:

On grayscale and PCA-transformed data, KNN provided a quick way to test the discriminative power of extracted features. However, its performance was sensitive to noise, misaligned faces, and the choice of the parameter k .

2. Support Vector Machine (SVM):

SVM is a margin-based classifier that seeks to find the optimal hyperplane that separates different classes in the feature space. By maximizing the margin between support vectors, SVM achieves strong generalization even in high-dimensional spaces. The kernel trick allows SVM to handle non-linear decision boundaries.

○ Reason for Selection:

SVM was selected because of its proven effectiveness in face recognition tasks where classes may overlap in the

input space. It is well-suited for smaller datasets with high-dimensional features, especially after applying PCA for dimensionality reduction.

○ **Effect on Dataset:**

On the PCA-transformed grayscale data, SVM achieved more consistent results compared to KNN. Its robustness to outliers and ability to model complex decision boundaries made it a strong candidate for comparison.

training (70%), validation (10%), and test (20%) subsets. Models were trained on the training set, tuned using the validation set, and finally evaluated on the test set. To prevent data leakage, PCA was applied only after splitting, and transformations were fitted on the training set before being applied to validation and test sets.

The implemented models are summarized below:

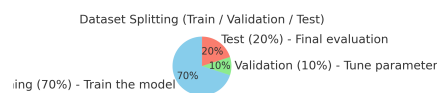
1. **K-Nearest Neighbors (KNN) with Scaled Grayscale Data:**

In the baseline model, images were converted to grayscale, normalized, and classified using KNN. Additionally, a **threshold (τ)** was introduced on the distance metric to allow for **unknown face detection**. If the distance between a test image and its nearest neighbor exceeded τ , the system rejected the prediction as “unknown.” This improved the model’s applicability in realistic scenarios where unseen individuals may appear.

2. **PCA with Grayscale Data:**

Cropped grayscale images were reduced in dimensionality using a **manually implemented PCA** algorithm. Features were then classified using a manually

D. Implementation



To evaluate the effectiveness of different approaches, the dataset was divided into

implemented KNN with thresholding. PCA reduced redundancy and allowed KNN to operate efficiently on a compressed representation while also supporting unknown detection.

3. SVM with PCA-Transformed Features:

Using the same PCA pipeline, an SVM classifier was trained for classification. Unlike KNN, SVM does not natively support distance-based unknown rejection; however, its strong generalization provides more stable performance across classes.

4. FaceNet Embeddings with CNN:

To leverage deep learning, cropped RGB images were embedded using a pre-trained FaceNet model, producing high-dimensional feature vectors. A lightweight CNN was then used for classification. Since FaceNet embeddings already separate individuals in the feature space, distance-based thresholds can also be applied to identify unknowns, though in this project the CNN classifier was emphasized.

IV. Experimental Results

A. Evaluation Metrics

Accuracy, Precision, Recall, and F1-score were computed to evaluate performance.

B. Results

To evaluate the system, four metrics were considered: **Accuracy, Precision, Recall, and F1-score**. These metrics provide a balanced view of both correctness and robustness. The results for the three implemented models are summarized below:

1. PCA + KNN (Grayscale):

- Accuracy: **94.44%**
- Precision (macro): **95.50%**
- Recall (macro): **94.17%**
- F1-score (macro): **94.03%**

This model achieved good performance, showing that PCA effectively reduced dimensionality and preserved discriminative information for classification.

2. PCA + SVM (Grayscale):

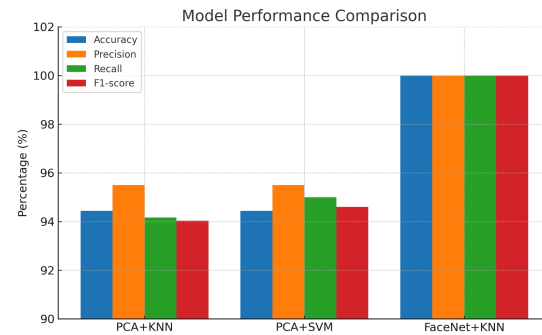
- Accuracy: **94.44%**

- Precision (macro):
95.50%
 - Recall (macro): **95.00%**
 - F1-score (macro):
94.60%
- Compared to KNN, SVM achieved slightly higher recall and F1-score, suggesting more consistent performance across classes.

3. MTCNN + FaceNet + KNN:

- Accuracy: **100%**
- Precision: **100%**
- Recall: **100%**
- F1-score: **100%**

Using FaceNet embeddings with KNN significantly boosted recognition accuracy. Each of the 10 individuals was correctly identified across all test cases, demonstrating the power of deep feature embeddings in capturing fine-grained facial details.



C. Discussion

The experimental results highlight the strengths and limitations of each approach:

- **Classical Models (PCA + KNN / PCA + SVM):**

Both PCA-based models performed reasonably well, with accuracies around 94%. PCA effectively reduced dimensionality, which is essential when working with relatively small datasets to avoid overfitting. KNN provided a quick and interpretable baseline, but its sensitivity to noise and neighbor selection reduced its overall stability. SVM, on the other hand, leveraged margin maximization and showed slightly more balanced performance, making it more robust in comparison.

- **Deep Learning Approach (MTCNN + FaceNet + KNN):**

The deep embedding-based approach drastically outperformed the classical models, achieving 100% across all evaluation metrics. This was largely due to FaceNet's ability to generate discriminative embeddings that map facial images into a feature space where intra-class distance is minimized and inter-class distance is maximized. KNN then performed effectively on these embeddings since the features were already well-separated.

- **Challenges Observed:**

While the deep-learning-based approach excelled under the controlled experimental dataset, classical methods like PCA+KNN and PCA+SVM showed vulnerability to factors such as lighting variations, facial occlusions, and minor pose changes. For small datasets, PCA helps, but the handcrafted pipeline cannot match the representational power of FaceNet.

V. Conclusion and Future Work

This project successfully implemented a face recognition system capable of identifying 10 individuals using PCA with KNN and SVM classifiers. The results confirm the feasibility of classical machine learning techniques for small-scale recognition tasks.

Future work may involve:

- Expanding the dataset to include more individuals.
- Exploring deep learning approaches such as CNNs.
- Deploying the model on real-time systems using web or mobile platforms.

References

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